Karthikeyan Sethuraman

ksethuraman1992@gmail.com|Linkedin| (732) 532-9389 |Plano, TX

SUMMARY

Data Science professional with 7 years of work experience in marketing, retail and finance (credit risk) domains with strong interpersonal skills. Proficient in machine learning, statistical modelling, and software programming.

EDUCATION

Rutgers University, New Brunswick, New Jersey **Dec 2020**
Master of Science in Data Science - Statistics Track – GPA - 3.75

Coursework: *Data Mining and Machine Learning, Regression and Time Series, Database systems, Statistical Inference*

National Institute of Technology, Trichy, India
Bachelor of Technology – Industrial Engineering **–** GPA - 3.7 **May 2014**

TECHNICAL SUMMARY

Languages: Python, R, SAS, SQL, Java **Databases:** Teradata, Oracle SQL, MySQL, Apache Spark

Core Domain Expertise: Supervised Learning (Regression-Linear, Ridge, Lasso, Logistic | Decision Trees | SVM | Neural Networks, Random Forest | Gradient Boosting), Unsupervised Learning (K-Means | Dimensionality Reduction | PCA | Cluster Analysis), ANOVA, Deep Learning (CNN | RNN), NLP, Ensemble Learning, Regularization, ML Interpretability

Analytics skillset: Speed Forecasting, Loan Origination Scorecards, Expected Credit Losses Modelling, Recommendation Systems, A/B testing, Market mix modelling, Churn Analysis, RFM Segmentation, Cross-sell Models

PROFESSIONAL EXPERIENCE

Senior Data Scientist, **CVS Health,** Plano, TX, USA **Mar 2023 – Present**
Risk Adjustment Product
- Developing a new health risk adjustment product for Medicare customers that identifies risk coding gaps and optimises member outreach with multi channel engagement to connect members with next best action for health screenings to close gaps.

Data Scientist, **Amazon,** Bellevue, WA, USA **Aug 2021 – Mar 2023**
Speed Forecasting | Python,
- Designed and automated a weekly Glance View Speed Forecasting and reporting framework for the transportation execution team that predicts the network speed at various levels based on the inventory placement and transportation capacity data.
- Built various Long-term forecasting Models to provide quarterly guidance on various speed related metrics to the finance stakeholders.

Assistant Manager, **Genpact,** Plano, TX, USA **Feb 2020 – Aug 2021**
Risk Data Science | Python
- Built a universal ***Market value model*** for an auto-finance client, using gradient boosting, to provide a market value forecast of a car at any point in time throughout its lifecycle for residual value setting and remarketing.
- Developed an ***Expected Credit Losses*** Framework to forecast probability of default and early payoff for consumer loans. The framework also automatically evaluates the model's continuing performance through backtesting.

Senior Data Analyst, **Epsilon,** Bangalore, India **Aug 2017 – Jul 2018**
Retail Analytics | R, Python, SAS
- Built a churn model for a retail client using logistic regression (AUC - 0.73) to predict the customers likely to unsubscribe from marketing mails. Enhanced the model using gradient boosting to get a top decile lift of 8.5.

Senior Data Analyst, **Citibank,** Chennai, India **May 2016 – Aug 2017**
Credit cards and Retail Banking Analytics | SAS, Teradata, Excel
- Increased the campaign response rate by ~22% for an insurance product, by building a ***cross-sell acquisition model*** using. logistic regression to predict the retail banking customers who are most likely to sign up.
- Led a 5-member team that collaborated with the marketing team to design and track various Acquisition, Portfolio. Engagement, Retention, Service, Cross-sell campaigns yielding ~ 1-2 % incremental transactions in the Hong Kong market.

Data Analyst, **PayPal(LatentView Analytics),** Chennai, India **Jun 2014 – May 2016**
Campaign Analytics | R, SAS, SQL Server, Teradata, Tableau
- Designed A/B testing for several marketing campaigns for one of the largest payments industries in the United States.
- Generated a synthetic control group using ***propensity score matching*** (logistic regression) with a target group. Difference of difference analysis of the two groups was done to study the impact of a product on customer spend.

ACADEMIC PROJECT – *IN GITHUB*

- **Content-based recommendation system for New York Times articles** **April 2019**
  Developed an application that enables user to input a set of keywords and uses Spark RDDs to retrieve relevant articles from the local database (NY Times articles) and computes text similarities based on TF-IDF, Jaccard and Cosine Similarity.